Day 9 — TNPSC Group 2A Study Material

Part A – General Studies: Geography of India: Location & Physical Features

READING MATERIAL

This is the first of Unit II's sub-topics, covering India's location, landforms, plateaus, rivers, and borders. Real exam evidence favors statement-evaluation ('which of these is/are correct') and superlative facts (longest, oldest, biggest) over plain description.

Q: Real exam: Evaluate these statements about the Malwa Plateau — (1) average altitude 600m, slopes east (2) mostly drained by the Chambal and its tributaries (3) located in Rajasthan, Gujarat, and Madhya Pradesh.

A: (2) and (3) are correct; (1) is false (the altitude/slope-direction claim doesn't hold). Malwa Plateau is genuinely drained by the Chambal river system and spans Rajasthan, Gujarat, and Madhya Pradesh.

Q: Real exam: Evaluate these statements about Indian soils — (1) northern plains' soils are largely deposited by Himalayan rivers (2) iron oxide presence is more characteristic of black soil (3) Deccan-trap-derived soil is fertile with high moisture-holding capacity.

A: (1) and (3) are correct; (2) is false — high iron oxide content is actually characteristic of RED soil (which gets its colour from iron oxide), not black soil. Black soil (regur) is derived from basalt/Deccan trap, is clay-rich, and is well suited to cotton cultivation.

Q: Real exam: What is the northern upland of the Karnataka Plateau called?

A: Malnad.

Q: Real exam: What is the largest west-flowing river of peninsular India?

A: The Narmada (not Mahi, Sabarmati, or Luni).

Q: What is the height of Anaimudi (the highest peak of the Peninsular Plateau / Western Ghats) above mean sea level?

A: 2,695 m.

Q: What is the approximate length of the Himalayan range, and which mountain system in India is the OLDEST?

A: Himalayan range length ≈ 2,560 km. The Aravalli Range is the oldest Indian mountain system (older than the Himalayas, Siwaliks, or Western Ghats).

Q: With which neighbouring country does India share its LONGEST international border?

A: Bangladesh.

Q: Which country in the world shares a border with the maximum number of other countries?

A: Russia.

Q: Which is India's first Island district?

A: Majuli (a river island in the Brahmaputra, Assam — notable as a district that is also a river island, distinct from the Andaman & Nicobar or Lakshadweep island UTs).

Q: Arrange these Himalayan mountain ranges from north to south: Karakoram, Ladakh, Pir-Panjal, Zaskar.

A: Karakoram → Ladakh → Zaskar → Pir-Panjal (north to south).

MOCK TEST (WITH ANSWERS)

1. The Malwa Plateau is mostly drained by which river and its tributaries?

A. Narmada

B. Chambal ✓

C. Tapi

D. Mahi

Explanation: The Chambal river system. [medium, question-bank-2025]

2. Which colour of soil is most associated with high iron oxide content?

A. Black soil

B. Red soil ✓

C. Alluvial soil

D. Laterite soil only in patches

Explanation: Red soil gets its colour from iron oxide content. [medium, question-bank-2025]

3. Soil derived from the Deccan trap is known for:

A. Poor fertility and low moisture retention

B. High fertility and high moisture-holding capacity ✓

C. Being entirely sandy

D. Being unsuitable for any cultivation

Explanation: Deccan trap-derived (black/regur) soil is fertile with high moisture retention. [medium, question-bank-2025]

4. The northern upland of the Karnataka Plateau is known as:

A. Malnad ✓

B. Maidan

C. Betwa

D. Dhasan

Explanation: Malnad. [medium, question-bank-2024]

5. The largest west-flowing river of peninsular India is the:

A. Mahi

B. Sabarmati

C. Narmada ✓

D. Luni

Explanation: The Narmada. [easy, question-bank-2022]

6. The height of Anaimudi above mean sea level is approximately:

A. 2,540 m

B. 2,455 m

C. 2,695 m ✓

D. 2,715 m

Explanation: 2,695 m — the highest peak of the Peninsular Plateau. [medium, web-research]

7. Which is the oldest mountain system in India?

A. Siwalik Range

B. Aravalli Range ✓

C. Western Ghats

D. Vindhya Range

Explanation: The Aravalli Range. [medium, web-research]

8. India shares its longest international border with:

A. Pakistan

B. Nepal

C. Bangladesh ✓

D. Myanmar

Explanation: Bangladesh. [medium, web-research]

9. Which country shares a border with the maximum number of other countries?

A. China

B. Russia ✓

C. India

D. Brazil

Explanation: Russia. [medium, web-research]

10. Which is India's first Island district?

A. Majuli ✓

B. Andaman

C. Diu

D. Lakshadweep

Explanation: Majuli, a river island district in the Brahmaputra (Assam). [hard, web-research]

Part B – Aptitude & Mental Ability: Volume

READING MATERIAL

Volume is the last of 10 named Aptitude sub-skills. Real exam evidence shows two dominant shapes: 'ratio of dimensions → ratio of volumes' (scaling), and 'melt/recast one solid into another shape' (volume is conserved, surface area is not) — both are core techniques worth knowing cold.

Q: Method — ratio of volumes from a ratio of radii (same shape): If the radius of one sphere is half that of another, what is the ratio of their volumes?

A: Volume scales with the CUBE of linear dimensions ($V=4/3\pi r^3$), not directly with the radius. So if $r_1:r_2=1:2$, then $V_1:V_2 = 1^3:2^3 = 1:8$.

Q: Method — melting and recasting (volume is conserved, NOT surface area): A sphere of radius 9cm is melted and recast into a cone of the same radius (9cm). Find the cone's height.

A: Set sphere volume = cone volume: $(4/3)\pi(9)^3 = (1/3)\pi(9)^2h \rightarrow (4/3)(729) = (1/3)(81)h \rightarrow 972 = 27h \rightarrow h = 36$ cm. The general shortcut: for melting sphere→cone of the SAME radius, height = 4×radius always (since the $(4/3)r^3$ vs $(1/3)r^2h$ relationship reduces to $h=4r$ when radii match).

Q: Real exam (already seen in an earlier topic, re-noted here): A sphere of diameter 8cm is melted and drawn into a wire of diameter 3mm. Find the wire's length.

A: Volume conserved: $(4/3)\pi(4)^3 = \pi(0.15)^2\times h \rightarrow h \approx 3792.6$ cm ≈ 37.9 m. (Same melt/recast principle as r02, just sphere→cylinder/wire instead of sphere→cone.)

Q: Method — volume of a cylindrical tank from its capacity: A cylindrical tank has capacity 1,848 m^3 and base diameter 14m. Find its depth.

A: $V=\pi r^2h \rightarrow 1848 = (22/7)\times7^2\times h = 154h \rightarrow h = 12$ m.

Q: Method — dug-out soil redistributed into a different shape (volume conserved): A cylindrical well of depth 20m, diameter 14m is dug; the soil is spread evenly to form a cuboid platform of base 20m×14m. Find the platform's height.

A: Well volume = $(22/7)\times7^2\times20 = 3{,}080$ m^3. Platform volume = $20\times14\times h = 280h$. Set equal: $280h=3080 \rightarrow h=11$ m.

Q: Formula check: volume of a cube with side 5cm, and the relationship between a cube's volume and surface area.

A: Volume = $\text{side}^3 = 5^3 = 125$ cm^3. If a cube's volume is 1000 cm^3, its side = $\sqrt[3]{1000}$ = 10cm, so its surface area = $6\times\text{side}^2 = 6\times100 = 600$ cm^2 (cube surface area = $6\times\text{side}^2$, from its 6 square faces).

Q: Method — ratio of volumes for cone, sphere, cylinder of equal radius AND equal height: What is that ratio?

A: Cone : Sphere : Cylinder = 1 : 4 : 3, when all three share the same radius r and the same height h (for the sphere, 'height' = its diameter = 2r in this comparison). This is a frequently tested ratio worth memorizing directly.

MOCK TEST (WITH ANSWERS)

1. If the radius of sphere A is 3 times the radius of sphere B, what is the ratio of their volumes (A:B)?

A. 3:1
B. 9:1
C. 27:1 ✓
D. 6:1

Explanation: Volume ratio = (radius ratio)^3 = $3^3:1^3$ = 27:1. [easy, authored]

2. A solid sphere of radius 6cm is melted and recast into a cone of the same radius. Find the cone's height.

A. 18 cm
B. 24 cm ✓
C. 12 cm
D. 6 cm

Explanation: Using the shortcut h=4×radius for sphere→same-radius-cone: h=4×6=24cm. [medium, authored]

3. A cylindrical tank has capacity 4,620 m^3 and base diameter 14m. Find its depth.

A. 20 m
B. 25 m
C. 30 m ✓

D. 15 m

Explanation: $V=\pi r^2h \rightarrow 4620 = (22/7)\times 49\times h = 154h \rightarrow h=30$m. [medium, authored]

4. Find the volume of a cube with side 8cm.

A. 512 cm³ ✓

B. 256 cm³

C. 64 cm³

D. 640 cm³

Explanation: $8^3 = 512$ cm³. [easy, authored]

5. If a cube's volume is 2,197 cm³, find its surface area.

A. 507 cm²

B. 169 cm²

C. 1014 cm² ✓

D. 338 cm²

Explanation: Side = $\sqrt[3]{2197} = 13$cm. Surface area = $6\times 13^2 = 6\times 169 = 1{,}014$ cm². [medium, authored]

6. A cone, a sphere, and a cylinder share the same radius and same height. What is the ratio of their volumes (cone:sphere:cylinder)?

A. 1:3:4

B. 1:4:3 ✓

C. 3:4:1

D. 4:3:1

Explanation: Cone:Sphere:Cylinder = 1:4:3 for equal radius and height. [medium, web-research]

Part C – General English: Poem: "Your Space" — David Bates

READING MATERIAL

A didactic poem urging gentle, kind speech toward everyone — especially children, the elderly, and the poor. Real coaching-site evidence tests theme, in-context word meaning, and figure-of-speech together, similar to the other Guest/Van-Dyke-style moral poems already covered.

Q: What is the central theme/message of "Your Space"?

A: Speak gently and kindly — to everyone, but especially to children, the elderly, and the poor. Harsh words can ruin the good we might otherwise do; kindness builds trust and love far better than fear or force does.

Q: Why does the poet say we should speak gently to children specifically?

A: To earn their trust and love — childhood is brief, and gentle, caring words said to children in that short period are the ones they'll remember always.

Q: Why does the poet urge gentle speech toward the elderly?

A: Because in their last years, all they need is affection and love — the poet says to speak to them as tenderly as to a child, so they may depart in peace.

Q: Why does the poet urge gentle speech toward the poor?

A: The poor are already suffering in an unkind world — there's no need to add to that suffering with harsh tones or unkind words.

Q: What does "Sands of life are nearly run" refer to?

A: Time running out for the aged — an image of an hourglass nearly empty, used to describe someone near the end of their life.

Q: What does "mar" mean in "let not harsh words mar the good we might do here"?

A: Ruin/spoil — harsh words can ruin good deeds, even when the deeds themselves were well-intentioned.

Q: Identify the figure of speech in "Speak gently! – 'tis a little thing / Dropped in the heart's deep well."

A: Metaphor — a spoken word is described AS something physically 'dropped' into the heart, pictured as a 'deep well', without using 'like/as'.

MOCK TEST (WITH ANSWERS)

1. "Your Space" was written by:

A. Rudyard Kipling

B. David Bates ✓

C. Sarojini Naidu

D. Robert Frost

Explanation: David Bates. [easy, web-research]

2. The main theme of the poem is:

A. Anger and revenge

B. Speaking powerfully

C. Speaking gently and kindly ✓

D. Arguing with boldness

Explanation: Speaking gently and kindly to everyone. [easy, web-research]

3. In the poem, why should we speak gently to a child?

A. To avoid punishment

B. To earn their trust and love ✓

C. To teach discipline

D. To save time

Explanation: To earn their trust and love. [medium, web-research]

4. "Sands of life are nearly run" refers to:

A. A desert trip

B. Time running out for the aged ✓

C. A sand-clock game

D. A beach adventure

Explanation: Time running out for the aged. [medium, web-research]

5. In "let not harsh words mar the good we might do here," the word "mar" means:

A. Help

B. Fix

C. Ruin ✓

D. Repair

Explanation: Ruin/spoil. [medium, web-research]

6. "Speak gently! – 'tis a little thing / Dropped in the heart's deep well" uses which figure of speech?

A. Simile

B. Personification

C. Metaphor ✓

D. Alliteration

Explanation: Metaphor — a spoken word equated with something dropped into a 'deep well'. [medium, web-research]